GIS Wetland Degradation & Depletion Analysis

Multi-Spectral Remote Sensing & Multi-Evidence Change Assessment Report

Run ID:	run_20260829_115448	**Baseline (T1):**	2023-08-14
Software:	GIS Engine v1.0.0	**Present (T2):**	2023-08-28
CRS:	EPSG:32633	**Interval:**	14 days
Resolution:	10.0 m × 10.0 m	**Analysis Area:**	36.00 ha

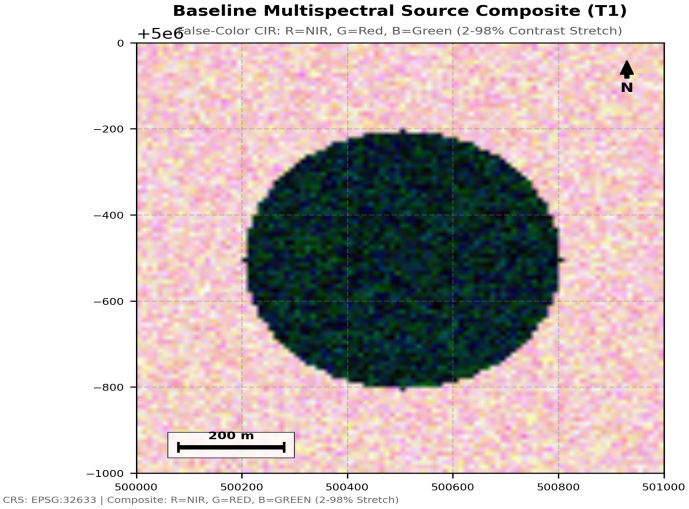
1. Executive Summary & Core Metrics

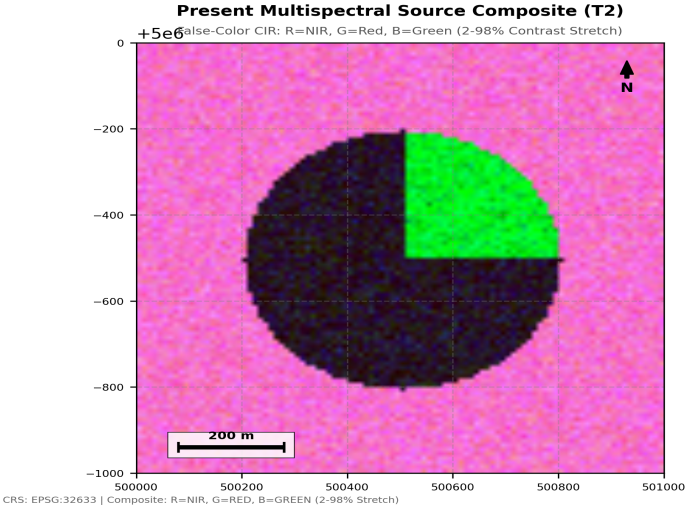
This technical assessment evaluates multi-spectral landscape and vegetation dynamics between **2023-08-14** (T1) and **2023-08-28** (T2) across a valid footprint of **36.00 ha**. Analysis integrates 5-channel ResNet34 U-Net inference, 16-state post-classification transitions, continuous 3D Change Vector Analysis (CVA), and McFeeters NDWI surface water modeling. Vegetation degradation was measured across **0.00 ha (0.00%)**, recovery across **0.00 ha (0.00%)**, and stable landscape across **100.00 ha (100.00%)**. The Severity-Weighted Degradation footprint is **0.00 ha** (NDR: **0.0000**). **Physical Wetland Depletion Status:** NOT_ASSESSED.

Indicator / Metric	Area (ha)	Proportion (%)	Classification & Operational Scope
Vegetation Degradation	0.00	0.00%	Primary transitions: Healthy/Moderate → Severe, Healthy → Moderate
Vegetation Recovery	0.00	0.00%	Primary transitions: Severe/Moderate → Healthy, Severe → Moderate
Stable Landscape	100.00	100.00%	Persistent baseline state (H→H, M→M, S→S, B→B)
Severity-Weighted Footprint	0.00	-	Weighted aggregate: Σ (severity_weight × transition_area)
Baseline Water Extent (T1)	0.31 ha	-	Classified NDWI surface water presence at T1
Surface Water Dynamics	0.00 ha	-	Water Loss: 0.00 ha | Gain: 6.72 ha

2. Input Data Acquisition & Multispectral Imagery

Input datasets comprise calibrated multispectral raster bands (Green, Red, RedEdge, NIR). Because Blue bandpass is unavailable in this sensor configuration, source visualizations are rendered as standard Color-Infrared (CIR) false-color composites (Red = NIR, Green = Red, Blue = Green) with 2%–98% contrast stretching.



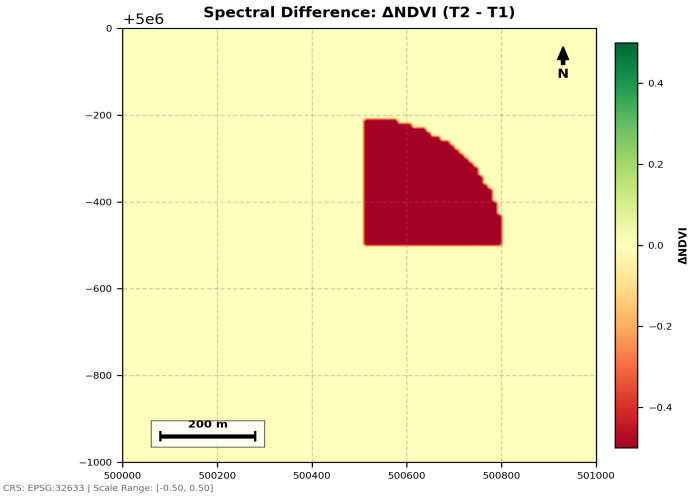


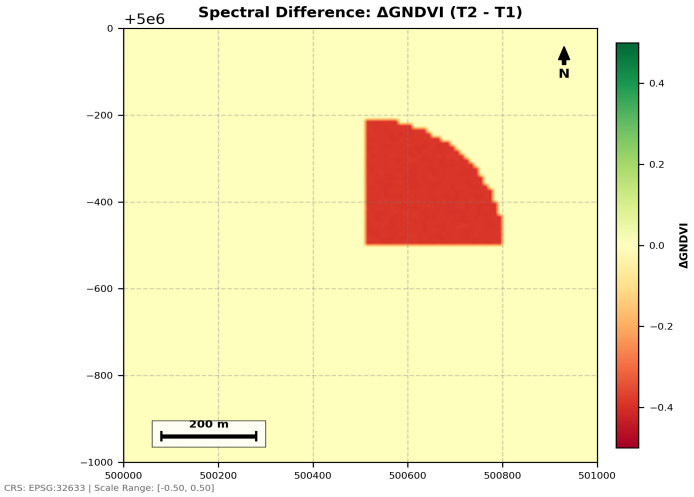
Band	Valid Pixels	T1 Min	T1 Max	T1 Mean	T1 Median	T2 Min	T2 Max	T2 Mean	T2 Median
Green	10,000	868.3	1,320.8	1,143.3	1,184.3	868.3	1,320.8	1,143.3	1,184.3
Red	10,000	483.4	924.5	744.2	784.9	502.3	1,234.9	776.6	792.8
RedEdge	10,000	926.8	2,777.8	2,177.6	2,574.7	573.4	2,777.8	2,147.7	2,574.7
NIR	10,000	912.2	5,146.2	3,783.9	4,759.0	443.3	5,146.2	3,739.4	4,759.0

3. Derived Spectral Indices & Continuous Trajectories

Authoritative vegetation and moisture indices are derived directly from the aligned multispectral raster bands. Statistics are computed strictly over the valid overlap mask.

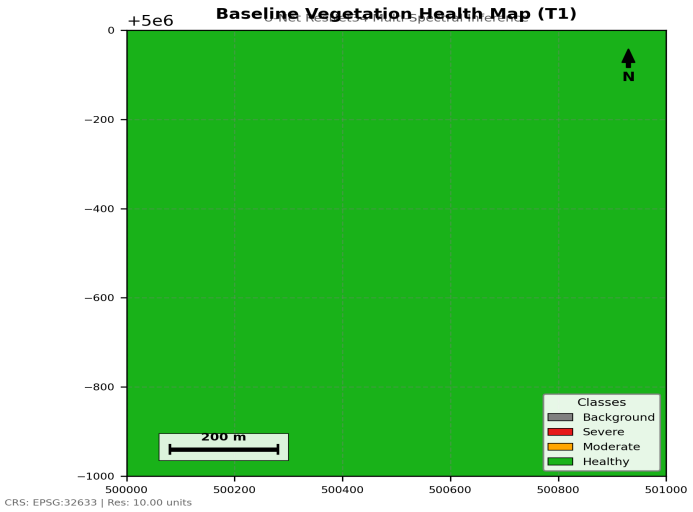
Index	Formulation	T1 Mean	T1 Median	T2 Mean	T2 Median	Mean Change (Δ)
NDVI	(NIR - Red) / (NIR + Red)	0.6061	0.7088	0.5611	0.7088	-0.0449
GNDVI	(NIR - Green) / (NIR + Green)	0.4559	0.5952	0.4296	0.5952	-0.0263
NDRE	(NIR - RedEdge) / (NIR + RedEdge)	0.2249	0.2913	0.2152	0.2913	-0.0097
NDWI	(Green - NIR) / (Green + NIR)	-0.4559	-0.5952	-0.4296	-0.5952	0.0263

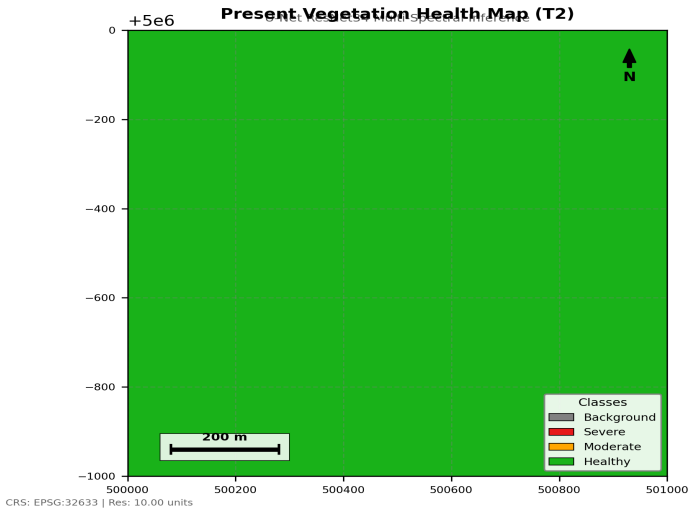




4. Vegetation Health & Spatial Condition (T1 vs. T2)

Vegetation health was independently inferred for T1 and T2 using the 5-channel ResNet34 U-Net model (NDVI, GNDVI, NDRE, NIR, RedEdge). Discrete classes: Background (0), Severe Stress (1), Moderate Stress (2), and Healthy Vegetation (3).

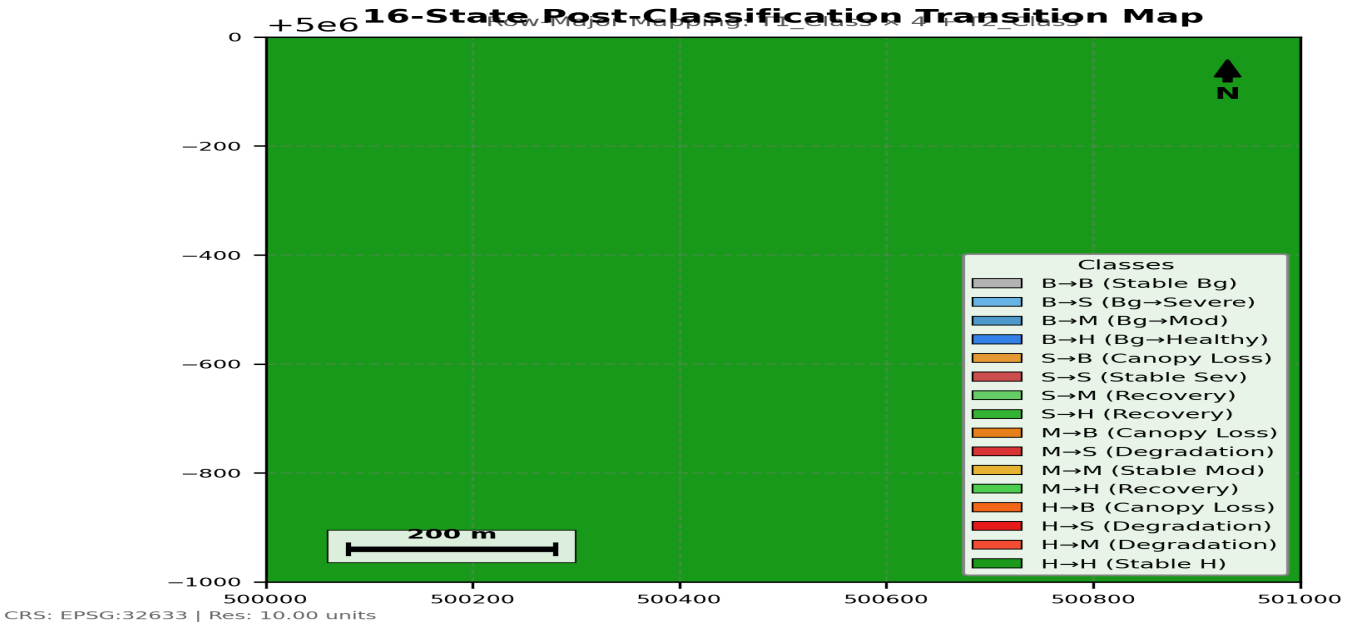




Vegetation Health Class	T1 Area (ha)	T1 Pct (%)	T2 Area (ha)	T2 Pct (%)	Net Shift (ha)
Healthy Vegetation	100.00	100.00%	100.00	100.00%	+0.00
Moderate Stress	0.00	0.00%	0.00	0.00%	+0.00
Severe Stress	0.00	0.00%	0.00	0.00%	+0.00
Background / Non-Veg	0.00	0.00%	0.00	0.00%	+0.00

5. 16-State Post-Classification Transition Analysis

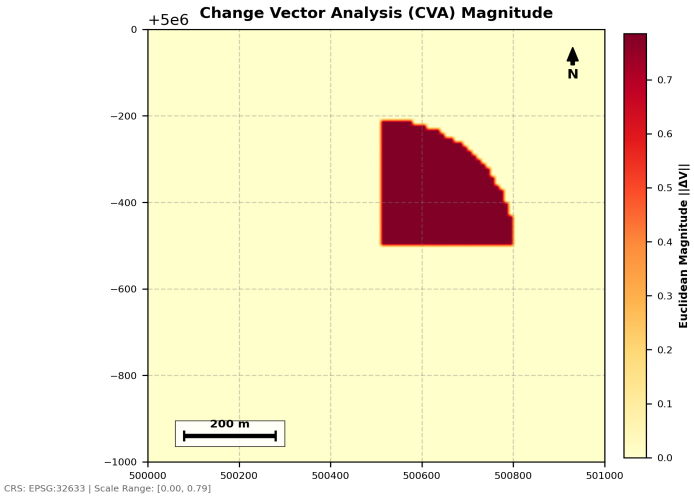
Cell-by-cell post-classification comparison maps temporal state transitions using deterministic row-major indexing (T1_class × 4 + T2_class). This isolates persistence from stress progression, canopy loss, and recovery dynamics.

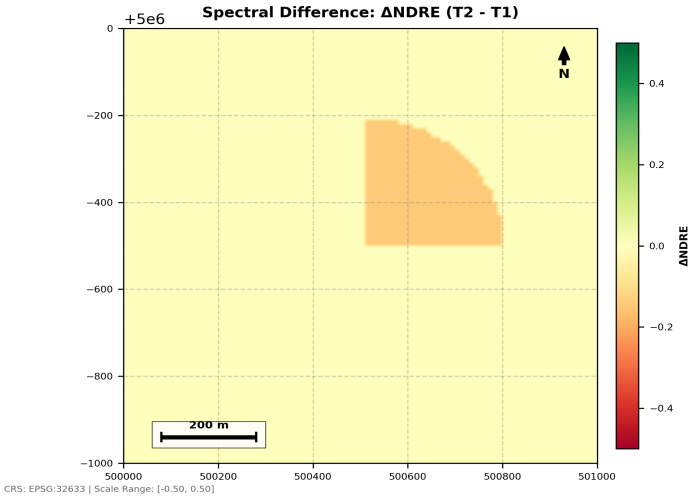


Transition Category	Transitions Included	Area (ha)	Percentage (%)
Stable State	H→H, M→M, S→S, B→B	100.00	100.00%
Vegetation Degradation	H→M, H→S, M→S	0.00	0.00%
Vegetation Recovery	M→H, S→M, S→H	0.00	0.00%
Vegetation → Background (Canopy Loss)	H→B, M→B, S→B	0.00	0.00%
Background → Vegetation (Emergence)	B→H, B→M, B→S	0.00	0.00%

6. Continuous Change Vector Analysis (CVA) & Residual Stress

Change Vector Analysis measures multi-spectral displacement across 3D vector space ([ΔNDVI, ΔGNDVI, ΔNDRE]). CVA magnitude captures subtle intra-class stress trajectories without double-counting discrete transitions.



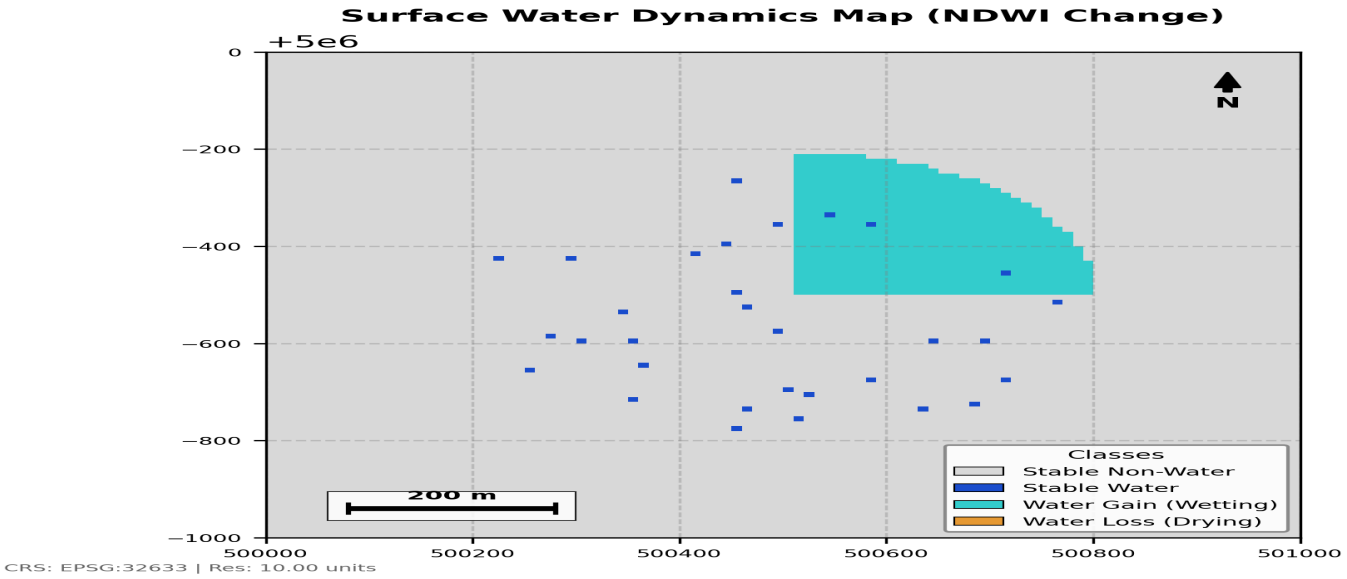


Spectral Metric	Mean Value	Median Value	95th Percentile	Physical Interpretation
CVA Magnitude (‖ΔV‖)	0.0529	0.0000	0.7837	Total spectral vector trajectory displacement
ΔNDVI (T2 - T1)	-0.0449	0.0000	0.0000	Photosynthetic vigor & chlorophyll absorption
ΔGNDVI (T2 - T1)	-0.0263	0.0000	0.0000	Green reflectance & canopy density

ΔNDRE (T2 - T1)	-0.0097	0.0000	0.0000	RedEdge boundary & foliar nitrogen stress

7. Hydrological Dynamics (NDWI & Surface Water)

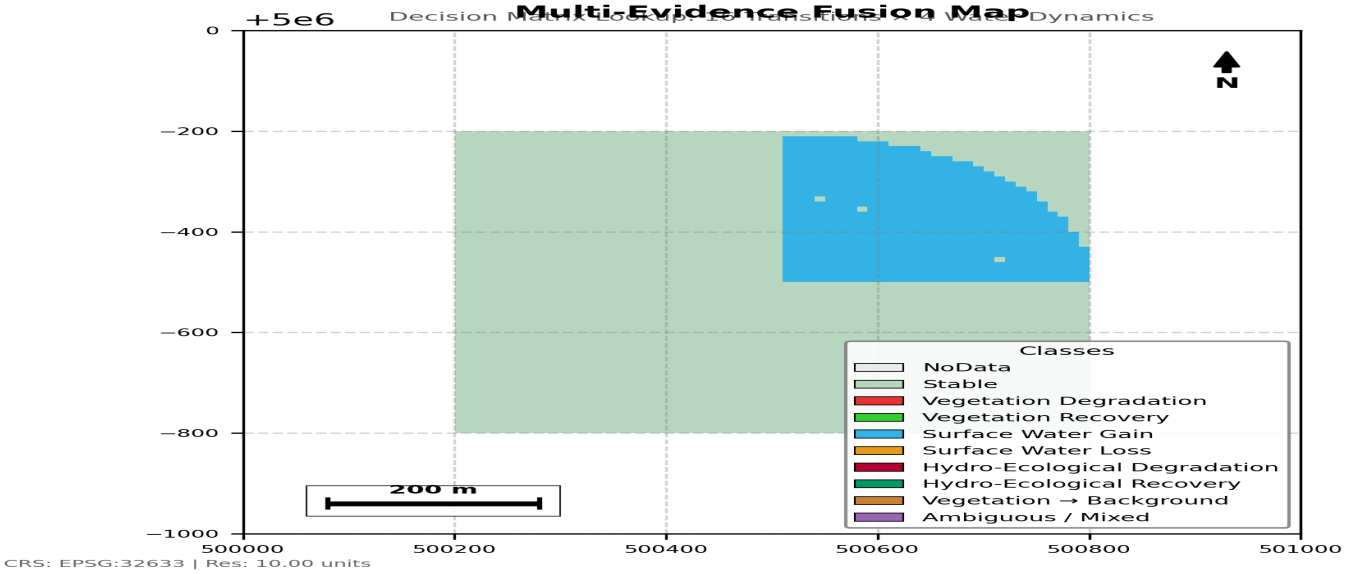
Surface water presence is modeled using the Green/NIR Normalized Difference Water Index (McFeeters 1996: [Green - NIR] / [Green + NIR]) with a baseline threshold of 0.0.



Hydrological Metric	Area (ha)	Hydrological State Definition
Baseline Water Extent (T1)	0.31 ha	T1 surface water area classified above NDWI threshold (2023-08-14)
Present Water Extent (T2)	7.03 ha	T2 surface water area classified above NDWI threshold (2023-08-28)
Surface Water Expansion (Gain)	6.72 ha	NDWI shifted below threshold to ≥ 0.0 (Flooding / Wetting)
Surface Water Contraction (Loss)	0.00 ha	NDWI shifted from ≥ 0.0 to below threshold (Drying / Receding)
NDWI Water Threshold	0.00	Standard McFeeters zero-plane threshold

8. Multi-Evidence Fusion Matrix

Multi-evidence fusion integrates vegetation health transitions with surface water dynamics using a 2D decision matrix (16 vegetation transitions × 4 water dynamics), separating true hydro-ecological decline from dryland vegetative stress.



Fusion Evidence Class	Area (ha)	Multi-Layer Evidence Definition
Stable Landscape	100.00 ha	Persistent baseline vegetation state and moisture regime

Hydro-Ecological Degradation	0.00 ha	Concurrent Vegetation Degradation + Surface Water Loss
Hydro-Ecological Recovery	0.00 ha	Concurrent Vegetation Recovery + Surface Water Gain
Vegetation-Only Degradation	0.00	Vegetation health decline occurring over stable water/moisture regime
Vegetation-to-Background (Canopy Loss)	0.00 ha	Vegetation loss without independent physical depletion confirmation

9. Quality Assurance & Validation Gates

Analytical steps are evaluated by automated Quality Assurance gates enforcing mathematical reconciliation, spatial grid integrity, and conservative scientific bounds.

Validation Check / Gate	Status	Quality Assurance Finding
Input Geometry & CRS Alignment	**PASS**	Exact grid match across dimensions, affine transform, and resolution.
Mathematical Area Reconciliation	**PASS**	Class areas (36.00 ha) reconcile with transition matrix sum (< 1e-4 ha residual).
Spectral CVA Distribution Check	**PASS**	No NaN/Inf leakage; finite bounds verified across all spectral bands.
Independent Ground Reference Data	**UNAVAILABLE**	Independent reference samples not configured. Areas represent model estimates.
Physical Depletion Claim Guard	**ENFORCED**	Safeguard active: Optical vegetation decline is not labeled physical wetland loss.

10. Scientific Interpretation & Limitations

OBSERVED (Direct Measurements): Pixel spectral reflectance values, band-derived indices (NDVI, GNDVI, NDRE, NDWI), CVA vector magnitude, and NDWI water plane shifts.
INFERRED (Multi-Layer Evidence): Vegetation health condition classes, canopy degradation/recovery transitions, and multi-evidence hydro-ecological stress.
NOT VALIDATED (Uncertainty Notice): Physical wetland loss/depletion cannot be confirmed without multi-year ground-truth validation or official cadastral wetland boundary records.

Technical Notice & Operational Bounds:
• **Independent Reference Data:** Ground reference validation samples are currently unavailable. Reported areas represent model-derived estimates rather than error-adjusted Olofsson confidence intervals.
• **Severity Weights:** Severity weighting factors (Healthy=0.0, Moderate=0.35, Severe=0.80, Background=1.0) are engineering calibration parameters and not field-calibrated ecological constants.
• **Sensor Bandpasses:** 4-band sensor data lacks SWIR; hence, MNDWI and LSWI formulations are excluded.
• **Phenological Dynamics:** Temporal gaps between acquisitions may reflect seasonal senescence rather than permanent anthropogenic degradation.

11. Technical Appendix: Data Provenance & Processing Trace

Traceability matrix mapping every primary report indicator back to its underlying raster artifact, calculation formula, and valid pixel counts.

Metric Name	Value	Unit	Source Artifact	Calculation / Method	Valid Count
Total Analysis Footprint	36.00	ha	alignment/overlap_mask.tif	Valid overlap pixel count × pixel_area_ha	3,600
Baseline Vegetated Area (T1)	100.00	ha	vegetation/t1_health.tif	ResNet34 U-Net Inference (Class 3)	10,000
Present Vegetated Area (T2)	100.00	ha	vegetation/t2_health.tif	ResNet34 U-Net Inference (Class 3)	10,000
Vegetation Degradation	0.00	ha	degradation/degradation_map.tif	Transitions (H→M, H→S, M→S)	0
Vegetation Recovery	0.00	ha	degradation/recovery_map.tif	Transitions (M→H, S→M, S→H)	0
Stable Landscape Area	100.00	ha	change/transition_map.tif	Transitions (H→H, M→M, S→S, B→B)	10,000
CVA Mean Magnitude	0.0529	index units	change/cva_magnitude.tif	Euclidean norm ‖[ΔNDVI, ΔGNDVI, ΔNDRE]‖	10,000
ΔNDVI Mean Difference	-0.0449	index units	change/delta_ndvi.tif	Arithmetic difference: NDVI(T2) – NDVI(T1)	10,000
Surface Water Extent (T1)	0.31	ha	hydrology/water_mask_t1.tif	McFeeters NDWI threshold: Green/NIR ≥ 0.0	31
Surface Water Loss	0.00	ha	hydrology/water_change.tif	NDWI transition: Water (T1) → Non-Water (T2)	-